Grounding Stress-Test Matrix

Sanity probes separate answer accuracy, control robustness, and explanation grounding.

One-glance insight: a model can reject blank images yet still rationalize wrong or mismatched visual evidence.

1. Probe outcomes

Green = expected grounding behavior observed, red = contradictory support, gray = heuristic unclear.

	Supports correct image Expected: support	Rejects wrong answer Expected: reject	Rejects blank image Expected: reject	Rejects shuffled image Expected: reject	Failure fingerprint Compact interpretation of the probe pattern.
Qwen2.5-VL-3B n = 20 probes per type Sanity source: spatial CoT	12/20 support 8/20 unclear	0/20 reject 16/20 contrad. 4/20 unclear	20/20 reject	3/20 reject 17/20 unclear	**Sees absence, misses mismatch** Detects blank inputs, but often accepts wrong answers or shuffled evidence.
DeepSeek-VL2 tiny n = 20 probes per type Sanity source: spatial CoT	0/20 support 20/20 unclear	3/20 reject 17/20 unclear	20/20 reject	20/20 reject	**Rejects controls, weak positive grounding** Flags blank and shuffled controls, but rarely phrases correct-image explanations as support.

● expected behavior observed ● contradictory support ● heuristic unclear

2. Misleading-question anomaly

A qualitative stress case for ambiguity and insufficient image grounding.

source_index 408 image context

Question prior

Modified prompt: pick up the object. The target is underspecified.

→

Model explanation

Assumes frying pan as the target and justifies the red arrow as motion toward it.

Interpretation

The failure mode is not merely an incorrect answer. The explanation can be coherent while being under-grounded: it follows a plausible linguistic prior instead of demonstrating that the target and arrow are visually verified in the scene.

Takeaway: accuracy, control robustness, and explanation grounding are separable axes. Sanity evaluations are most useful when they reveal which axis fails.

Generated from local sanity artifacts. Counts are heuristic labels over 20 selected examples per probe type.

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